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RESEARCH ARTICLE

Accelerated SPAD-Based Diffuse Optical Tomography With Data-Driven View Optimization

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ABSTRACT

Time-domain diffuse optical tomography (TD-DOT) is a non-invasive technique that utilizes near-infrared light to visualize the optical properties in tissues, showing promising applications in tumor diagnosis and functional brain imaging. The integration of advanced SPAD arrays delivers superior data quality characterized by picosecond timing precision and increased detection channels, enabling high-fidelity reconstruction of unmixed absorption and scattering properties. These gains, however, come with longer acquisition and computation times. We introduce a data-driven view-optimization strategy that exploits the statistical richness of time-resolved measurements to reduce the number of views without sacrificing image quality. We developed a noncontact TD-DOT combining a pulsed laser, a SPAD array, and a rotational stage for view selection. Simulations and phantom experiments show that the proposed method accelerates TD-DOT while maintaining robust reconstruction, comparable to those from full-view acquisition. The accelerated TD-DOT is valuable for monitoring biological dynamics, such as cerebral hemodynamics.

1 | Introduction

Diffuse optical tomography (DOT) is a non-invasive imaging technique that employs near-infrared light to probe the optical properties of living tissues in depth. The transmitted light from the boundary of tissues is captured by detectors and used to reconstruct the 3D map of the internal optical properties, that is, the absorption and the scattering coefficient. DOT reconstruction is known as an ill-posed inverse problem, which relies on a diffusive light propagation forward model. The reconstructed optical properties in tissues are important indicators of physiological and pathological conditions, for example, the absorption maps at multiple wavelengths are related to the concentration of deoxyhemoglobin and oxyhemoglobin [1]. Therefore, DOT constitutes a valuable tool in various areas, including tumor diagnostics [2], functional brain imaging [3, 4], and skeletal muscle assessment [5]. With different configurations of illumination and detection, DOT operates in three modes: continuous-wave (CW), frequency-domain (FD), and time-domain (TD). Among these modes, TD-DOT, which utilizes pulsed laser illumination and time-resolved detection, provides the richest measurement information, potentially enabling more accurate quantification of optical properties and improved depth sensitivity compared to the other two DOT modalities [6].

Abbreviations: CRLB, Cramér–Rao lower bound; CW, continuous wave; DOT, diffuse optical tomography; FD, frequency domain; FEM, finite element method; ICCD, intensified charge-coupled device; PDMS, polydimethylsiloxane; PMT, photomultiplier tube; SPAD, single-photon avalanche diode; SVD, singular value decomposition; TCSPC, time-correlated single photon counting; TD, time domain.

Early TD-DOT implementations generally fall into two categories: (1) contact systems based on optical fibers and photomultiplier tubes (PMTs) [7, 8] and (2) noncontact systems based on gated intensified charge-coupled devices (ICCDs) [9, 10]. The former provides high-accuracy and sensitive time-resolved data, but the combination of fibers and PMTs restricts the flexibility and detection channels, degrading reconstruction quality. The latter noncontact system is practical for both animal studies and clinical translation [11]. However, time-gated ICCD has relatively low temporal resolution (200 ps to 1 ns), limiting reconstruction quality and detection depth. Emerging single-photon avalanche diode (SPAD) arrays provide a compact, free-space alternative with superior data quality (up to 10 ps timing precision) and increasing number of channels [12, 13]. SPAD-based noncontact TD-DOT enables multi-view detection via rotational stages, crucial for high-resolution tissue absorption/scattering reconstruction [14]. However, the expanded data volume also extends acquisition and reconstruction times, posing challenges for real-time monitoring of transient biological events, such as hemodynamics. Optimizing projection views is thus essential for efficiency.

Optimization of projection views in a noncontact context is equivalent to optimizing the source-detector configuration in fiber-based contact systems [15]. Reported work on view optimization, or source-detector optimization, fall into two categories. The first category relies on the evaluation of the sensitivity matrix or Jacobian matrix of the forward problem. Xu et al. [16] introduced singular value decomposition (SVD) analysis to the sensitivity matrix in a simulation study of FD-DOT. Chen and Chen [17] used the Cramér–Rao Lower Bound (CRLB) to define the optimal precision for a given forward model and quantitatively estimated the reconstruction performance by testing various source-detector configurations. Additionally, Javidan et al. [18] developed an optimization framework based on the Kalman filter to find the optimal data acquisition strategy. The second category uses a data-driven strategy, mainly by assessing the correlation of the measurement data. Karkala and Yalavarthy [19] introduced a data-resolution matrix to assess the interdependence among different measurements, aiming to identify independent measurements for reconstruction. Shaw et al. [20] proposed a direct sensitivity matrix to select projection views with high signal-to-noise ratio (SNR). Overall, the first category necessitates the explicit calculation of the sensitivity matrix or Jacobian matrix, which is computationally expensive for TD-DOT in particular. More specifically, the accuracy of SVD analysis depends on threshold setting, while the estimation methods, such as evaluation via CRLB, tend to overlook practical challenges such as noise and the inaccuracy of forward modeling. The data-resolution matrix-based method requires heuristic determination of threshold which will influence the number of measurements. Currently, few studies have been done for assessing TD data [21]. The calculation of the Jacobian matrix contains the variable of time and incurs substantial computational cost, which is challenging for TD-DOT. To sum up, it is crucial to develop a highly efficient, robust and practical algorithm to optimize the projection views for accelerating TD-DOT.

We introduce a simple method that capitalizes on the rich statistical characteristics of TD measurements to optimize projection views in TD-DOT. We used the first moment derived from TD data and also analyzed its correlation and standard deviation, taking the influence of the data noise into account. We also developed a state-of-the-art noncontact TD-DOT system, equipped with a pulsed laser source and a SPAD array, and incorporated a rotational object stage to facilitate selective-view data acquisition. Through extensive numerical simulations and phantom studies, we demonstrate that our view optimization technique is fast, reliable, and effective for accelerating TD-DOT.

2 | Methods

2.1 | System Description

The schematics of the TD-DOT system are illustrated in Figure 1a. The imaging object is illuminated by a pulsed laser, and the transmitted light is captured by a SPAD array. A rotational stage (PRMTZ8/M, Thorlabs, USA) is adopted to facilitate free-space and multi-view acquisition. The light source is a picosecond pulsed laser with a repetition rate of 78 MHz (SuperK EXTREME, NKT, Denmark). The wavelength of the illumination light is set to 780 nm with a bandwidth of 10 nm by using an acousto-optic tunable filter (VARIA, NKT, Denmark). We employ a galvanometer scanner (basiCube 10, SCANLAB, Puchheim, Germany) to scan the sample surface with a lens (Lens 1, F290m, STARTNOW, Nanjing, China). The transmitted light is recorded by a SPAD array of 32×32 sensors [22] equipped with a focusing lens (Lens 2, ML183M, Theia Technologies, USA). An in-sensor time-correlated single photon counting (TCSPC) circuit within the SPAD sensor samples the measured temporal profile at each pixel with 256 time-bins and temporal resolution of 48.8 ps for each profile. The entire system is controlled by MATLAB, automating the whole data acquisition process, including scanning, stage rotation, and acquisition. The field of view (FOV) of the TD-DOT is approximately 50 mm × 50 mm. To block interference from ambient light, a homemade 3D-printed casing was used with an observation window of 30 mm×15 mm.

2.2 | Basic Principle of TD-DOT

Recovering optical properties from time-resolved measurement requires a high-precision simulation of diffuse light propagation in tissue, which is typically described by the diffusion equation (DE). Given a scattering medium domain $\mathbf{\Omega} \subset \mathbb{R}^3$, the DE in TD is expressed as follows [23]:

$$\left[\nabla \cdot \kappa(\boldsymbol{r})\nabla + \mu_a(\boldsymbol{r}) + \frac{1}{c}\frac{\partial}{\partial t}\right]\phi(\boldsymbol{r},t) = 0, \quad \boldsymbol{r} \in \mathbf{\Omega} \tag{1}$$

with a boundary condition:

$$\phi(\boldsymbol{m},t) + 2\zeta(c)\frac{\partial \phi(\boldsymbol{m},t)}{\partial \nu} = q(\boldsymbol{m},t), \quad \boldsymbol{m} \in \partial\mathbf{\Omega} \tag{2}$$

where $\phi(\boldsymbol{r},t)$ is the photon density distribution at position $\boldsymbol{r}$ and time t, and q is the source term. The important parameters of the model include the absorption coefficient μ_a, the diffusion coefficient $\kappa = \left[3\left(\mu_s'\right)\right]^{-1}$ with the reduced scattering coefficient μ_s', and the speed of light c. ζ is a constant accounting for the refractive index mismatch and $\partial\nu$ is an outward normal. The TD measurement is defined as [23]:

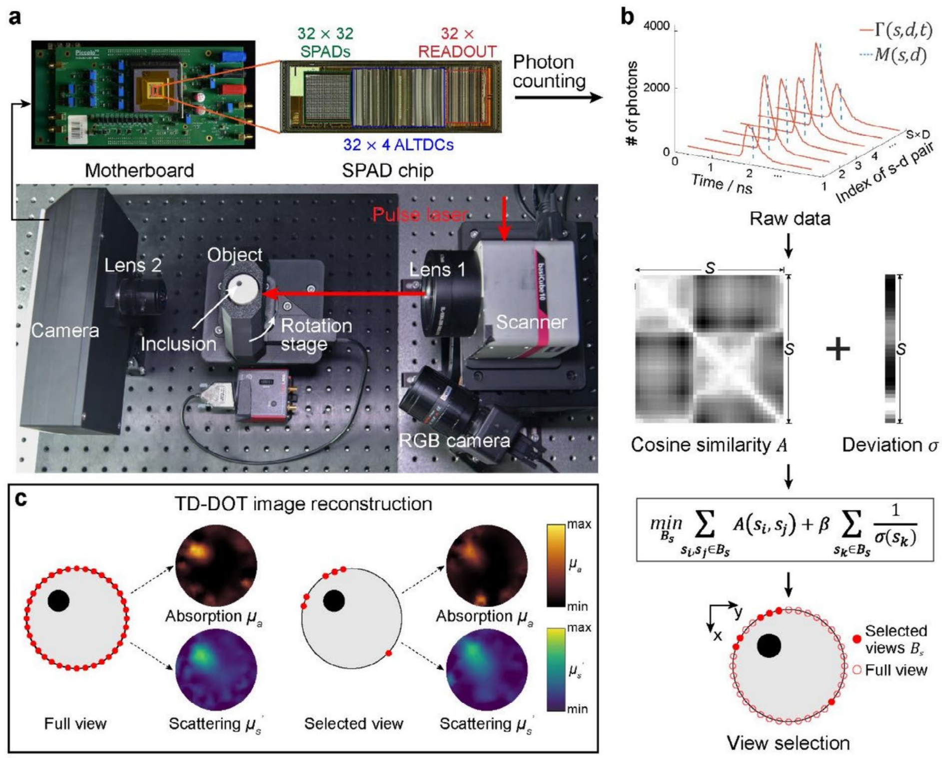

FIGURE 1 | TD-DOT setup and optimization of projection views. (a) The TD-DOT system integrates a pulsed laser, a time-resolved SPAD camera and a rotational stage for selective view acquisition. (b) The workflow of data-driven view optimization. (c) Comparison of the reconstruction results using full view and selected view B_S. The maps of μ_a and μ_s' reconstructed from the optimized views maintain image quality comparable to the full-view acquisition.

$$\Gamma(\boldsymbol{m}, t) = -\kappa(\boldsymbol{m}) \frac{\partial \phi(\boldsymbol{m}, t)}{\partial v} \tag{3}$$

To obtain $\phi(\boldsymbol{r}, t)$, finite element method (FEM) is used to solve DE numerically. NIRFAST [24], an FEM-based toolkit, is used in our DOT reconstruction. To reduce the computational cost, we convert TD data into FD via Fourier Transform and select the frequency component at 100 MHz for subsequent reconstruction [25].

To reconstruct absorption and scattering coefficients from boundary measurement, the following objective function Ψ should be minimized:

$$\Psi(\mu) = \left\| \Gamma_\omega^M - \Gamma_\omega^S(\mu) \right\|_2^2 + \lambda \left\| \mu - \mu_0 \right\|_2^2 \tag{4}$$

where the first term represents the difference between transformed experimental measurement Γ_ω^M and simulated FD measurement $\Gamma_\omega^S(\mu)$. $\mu = \left\{ \mu_a(\boldsymbol{r}), \mu_s'(\boldsymbol{r}) \right\}$ represents the optical properties to be determined and μ_0 is the initial guess. The second term is a Tikhonov-type regularizor [26] with a weighting parameter λ. For inversion, the algorithm updates optical properties following a Newton-type iterative process [24]:

$$\delta\mu = \left(J^T J + 2\lambda I \right)^{-1} J^T \delta\Gamma \tag{5}$$

where $\delta\mu$ is the update and $\delta\Gamma$ is the data-model misfit at the current iteration. The Jacobian matrix J defines the relationship between changes in simulated measurement Γ^S resulting from small changes in optical properties μ. The detailed construction of the Jacobian matrix can be found in [24].

2.3 | Data-Driven Optimization of Projection Views

Next, we investigate the optimization of the projection views to accelerate both the data acquisition and reconstruction. The measurement workflow two phases: (1) Initialization phase: this phase involves a single full-view scan (at $t = 0$) to compute the similarity matrix (A) and deviation vector (σ) of the full-view data, which determines the optimal views; (2) dynamic monitoring phase: this phase involves accelerated monitoring by applying the selected views, which is useful for monitoring the biological transients. Initially, we illuminate the object surface with a point-shaped laser beam and rotate the stage by a step size of Δs°, forming a full set of $S = 360 / \Delta s$ sources. For each source, D pixel-wise detectors and T discrete temporal bins are used. The full-view TD measurement is thus denoted as $\Gamma(s, d, t) \in \mathbb{R}^{S \times D \times T}$, corresponding to the measurement from the sth source and the dth detector at the time point of t (Figure 1b). From this initial full-view dataset, statistical features are extracted to determine an optimal subset of m views. For each source-detector pair, the first moment $M(s, d)$ is defined as

$$M(s, d) = \frac{\sum_{t=1}^{T} t\Gamma(s, d, t)}{\sum_{t=1}^{T} \Gamma(s, d, t)} \tag{6}$$

Next, we consider the influence of the data noise and evaluate the similarity between the measurements from different sources s_i and s_j using the metric of cosine similarity, forming a complete data correlation matrix $A \in \mathbb{R}^{S \times S}$:

$$A\left(s_i, s_j\right) = \frac{\sum_{k=1}^{D} M\left(s_i, k\right) M\left(s_j, k\right)}{\sqrt{\sum_{k=1}^{D} M\left(s_i, k\right)^2 \cdot \sum_{k=1}^{D} M\left(s_j, k\right)^2}} \tag{7}$$

Simultaneously, we calculate the standard deviation of the first moment-based data at the sth source, denoted as $\sigma \in \mathbb{R}^S$:

$$\sigma(s) = \sqrt{\frac{1}{D-1} \sum_{k=1}^{D} \left| M(s,k) - \frac{1}{D} \sum_{i=1}^{D} M(s,i) \right|^2} \tag{8}$$

For a free-space TD-DOT with a set of full projection views B_f, our goal is to select a set of "best" projection views B_s that contains m sources by minimizing the objective function:

$$f\left(B_s\right) = \sum_{s_i, s_j \in B_s} A\left(s_i, s_j\right) + \beta \sum_{s_k \in B_s} \frac{1}{\sigma\left(s_k\right)} \tag{9}$$

Equation (9) seeks a balance between spatial diversity and signal richness. Minimizing the similarity matrix A penalizes redundant measurements, essentially forcing the selected views to span a wide range of angles. Meanwhile, the deviation term σ acts as an indicator of signal strength. Prioritizing views with high variance ensures the algorithm captures the optical paths most heavily perturbed by internal inclusions. The hyperparameter β governs the trade-off between these constraints. In this study, β was tuned to 1×10^{-4}, where the competition between maximizing data diversity and capturing strong and independent signals forces the view configuration to expand its angular span. Detailed discussion for the tuning parameter β is provided in Section 1 of Supporting Information. To solve this objective function, we employ a greedy approach paired with a local search strategy, as outlined in Algorithm 1. Rather than an exhaustive search, the algorithm first builds an initial subset by iteratively adding the single view that minimizes the loss. To avoid getting trapped in local optima, it then swaps selected views with unselected ones, retaining any combination that further reduces the objective function. This iterative refinement efficiently approximates the optimal view configuration. The complete data-driven view optimization workflow is illustrated in Figure 1b, and a conceptual demonstration comparing full-view and selected-view reconstructions is shown in Figure 1c.

ALGORITHM 1 | Data-driven view optimization.

01: **Input:** B_f, m, A, σ
02: **Output:** B_s
03: **Initialization:** B_s starts with one random view
04: **while** $|B_s| < m$ **do**
05: Find a new view $b \in B_f$ to minimize $f\left(B_s\right)$
06: Add b to B_s
07: Remove b from B_f
08: **for** $i = 1, 2, \ldots, m$ **do**
09: $B_s' \leftarrow B_s$
10: **for** $j = 1, 2, \ldots, S - m$ **do**
11: $B_s'(i) \leftarrow B_f(j)$
12: **if** $f\left(B_s'\right) < f\left(B_s\right)$ **then**
13: $B_s \leftarrow B_s'$
14: **else**
15: Go to Step 8
16: **Return** B_s

2.4 | Numerical Simulations

We designed four numerical phantoms with varying geometries and inclusion distributions (Figure 2a). Cases 1–3 used a cylindrical phantom with a diameter of 30 mm and a height of 20 mm. Case 4 used a pillar-shaped phantom defined in a bounding box of $35 \times 23 \times 20\,\text{mm}^3$, for testing our method on irregular geometries. In all cases, the background optical properties were set to $\mu_a = 0.01\,\text{mm}^{-1}$ and $\mu_s' = 1\,\text{mm}^{-1}$. All target inclusions were modeled as cylinders (height: 8 mm, diameter: 6 mm) and centered at $z = 10\,\text{mm}$. The specific configurations were as follows: (1) a single inclusion with elevated contrast ($\mu_a = 0.03\,\text{mm}^{-1}$; $\mu_s' = 2\,\text{mm}^{-1}$) located at (10, 10, 10) mm; (2) two inclusions with identical high contrast, located at (10,10,10) and (20,10,10) mm; (3) two inclusions at the same locations as Case 2, but assigned with different optical properties to test their cross-talk: the upper inclusion features high absorption ($\mu_a = 0.03\,\text{mm}^{-1}$), and the lower inclusion features high scattering ($\mu_s' = 2\,\text{mm}^{-1}$); and (4) a single inclusion identical to Case 1 located at (8, 15, 10) mm within the pillar phantom.

During simulation, transmittance illumination was applied on the boundary at $z = 10\,\text{mm}$ with an interval of 10°. For each source, the central 18 detection pixels (6×3) with sufficiently high SNIR are used for reconstruction. The digital phantoms were discretized into tetrahedral meshes [27]. Among the simulated 36 views, we selected 6 views for TD-DOT reconstruction using different strategies: (1) our proposed data-driven method; (2) uniform down-sampling; and (3) SVD analysis of the Jacobian (J-SVD). For the J-SVD method, we adopted the singular-value analysis framework in [15, 28]. We constructed the FD Jacobian matrix J at 100 MHz and performed SVD on $J = U \Sigma V^T$. A greedy strategy was used to maximize the sum of the logarithmic singular values of the selected views. For comparison, reconstructions using the full-view dataset (36 views) were performed as the reference. Poisson noise was added in all simulation cases to mimic realistic experimental conditions, modeled with a peak photon count of ~10^3 yielding an average SNR of ~20 dB. DICE similarity coefficient (DSC) was used to assess the reconstruction quality with 55% thresholding. Furthermore, the mean absolute error (MAE) was calculated to quantify the pixel-wise accuracy of the reconstructed optical properties.

2.5 | Phantom Experiments

We conducted two phantom experiments (A and B) for static and dynamic imaging assessment, respectively. Both phantoms feature the identical cylindrical geometry (diameter: 30 mm; height: 50 mm). The optical properties were tuned by the mixing different concentration of titanium dioxide (Collins, CA, USA) and carbon powder (Collins, CA, USA), and calibrated via

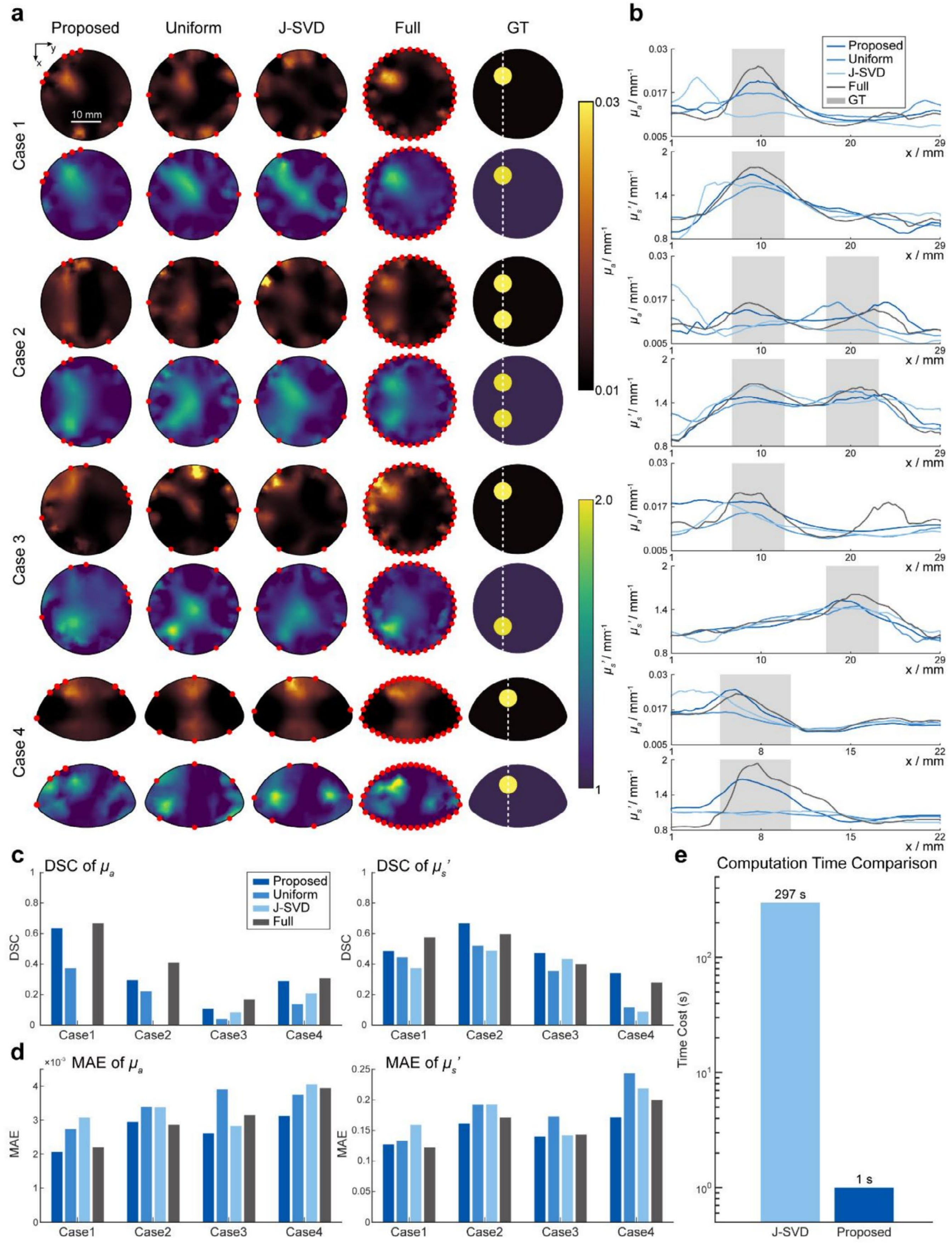

FIGURE 2 | Legend on next page.

FIGURE 2 | Comparison of reconstruction results using different view optimization methods across four simulation cases. (a) Reconstructed slices of absorption (μ_a) and scattering (μ_s'). Selected views using the proposed method, uniform down-sampling, and J-SVD are compared with the full-view reconstruction and the ground truth (GT). The selected views are marked with red dots. (b) Profile drawing obtained from the white dashed lines in (a). The gray shaded regions represent the locations of the inclusions in GT. (c) Quantitative evaluation using the DICE similarity coefficient for μ_a (left) and μ_s' (right) across all cases. (d) Quantitative evaluation using the mean absolute error (MAE) for μ_a (left) and μ_s' (right) across all cases. (e) Computational time comparison of view selection method. Scale bar: 10 mm.

TABLE 1 | Optical properties of the experiment material.

		μ_a (mm^{-1})	μ_s' (mm^{-1})
Background	Phantom A	0.01	0.76
	Phantom B	0.03	0.63
Inclusion	Solution A	0.04	2.5
	Solution B-1	0.017	0.42
	Solution B-2	0.13	2.65

integrating sphere-based measurement and the inverse adding-doubling method [29]. Phantom A contains a cylindrical cavity (diameter: 6 mm; depth: 35 mm) located 6 mm off the center. This cavity was filled with a mixture solution A of intralipid (MedChemExpress, USA) and India ink (Phygene, China). Phantom B contains a similar cylindrical cavity (diameter: 6 mm; depth: 27 mm). The cavity was initially filled with 400 μL mixture Solution B-1. During the data acquisition, another mixture Solution B-2 was continuously injected at a controlled rate of 100 μL/min using a micro-pump (PHD ULTRA, Harvard Apparatus, USA). The optical properties of the background and inclusions are summarized in Table 1.

The experimental data was acquired using a 780 nm pulsed laser. For the static experiment (Phantom A), a dataset covering all 36 views with a 10° interval with one laser point per view was acquired. The dynamic imaging experiment on Phantom B consists of three steps. First, the same full-view scan (36 views) was performed like the static imaging task. Second, dynamic imaging with six views picked with the proposed method was performed during injection Solution B-2, achieving six-times shorter acquisition time (~15 s per dataset, four datasets in total). Finally, a full 36-view scan was again performed for validation after injection.

3 | Results

3.1 | Results of Numerical Simulations

Comparison of the reconstruction results derived from different view selection methods are compared (Figure 2). The proposed data-driven method consistently achieves accurate reconstruction images closest to the full-view result, whereas other methods suffer from distinct image degradations. Specifically, in the single-target scenarios (Cases 1 and 4), the uniform method tends to produce reconstructions with blurred edges and high-intensity artifacts at the boundaries, while the results of J-SVD are compromised by distortions. These limitations are more apparent in complex settings. In the multi-target scenario (Case 2), uniform sampling and J-SVD fail to resolve both inclusions, merged into a single target. Conversely, our method successfully maintains spatial separation. In the cross-talk test (Case 3), while the uniform and J-SVD method manage to recover the scattering target, they fail to correctly reconstruct the absorption target. In contrast, our approach effectively decouples absorption from scattering perturbations. Quantitative evaluations (Figure 2b,c) corroborate these visual observations. Although all selected views exhibit some quantitative attenuation due to the reduced data fidelity term, our method preserves the structural fidelity more properly, achieving the highest DSC score. Furthermore, as demonstrated by the MAE evaluation (Figure 2d), the proposed method consistently achieves the lowest quantitative error, confirming its capability in accurate parameter recovery. Finally, the proposed method offers a significant efficiency advantage (Figure 2e). While J-SVD requires ~297 s, our method completes this task in less than 1 s, showing acceleration with over two orders of magnitude. Notably, the views selected by J-SVD and the uniform sampling should theoretically converge in symmetric objects (Cases 1–3), but, in reality, minor deviations of the view selection are found due to asymmetric discretization of meshing during FEM implementation (Figure 2a).

3.2 | Static Imaging Experiment

The results of static experiment are presented in Figure 3. The tomographic slices (Figure 3a) reveal that the proposed method achieves high reconstruction quality comparable with the full-view result. In the xy-plane at $z = 26$ mm, the inclusion recovered by our method exhibits sharp boundaries and high contrast, accurately locating the target. Notably, the yz-plane slices at $x = 15$ mm demonstrate that our method successfully reconstructs the vertical position with a precision comparable to the full-view result and the ground truth (GT) denoted by the dashed lines. It is worth noting since the ill-posedness of the sparse-view problem increases, all derived optical properties are slightly lower than those obtained from the full-view projection. Despite this attenuation, our method preserves the relative contrast and morphology significantly better than other methods. In contrast, the uniform sampling method yields a dispersed reconstruction with blurred edges and strong artifacts at boundaries. J-SVD produces a result similar to uniform sampling. Although it exhibits lower background artifacts, it still fails to clearly define the inclusion's boundaries, particularly in the vertical plane. Cross-sectional profiles (Figure 3b) further reveal that reconstruction using uniform sampling and J-SVD are flattened and broadened. The proposed method

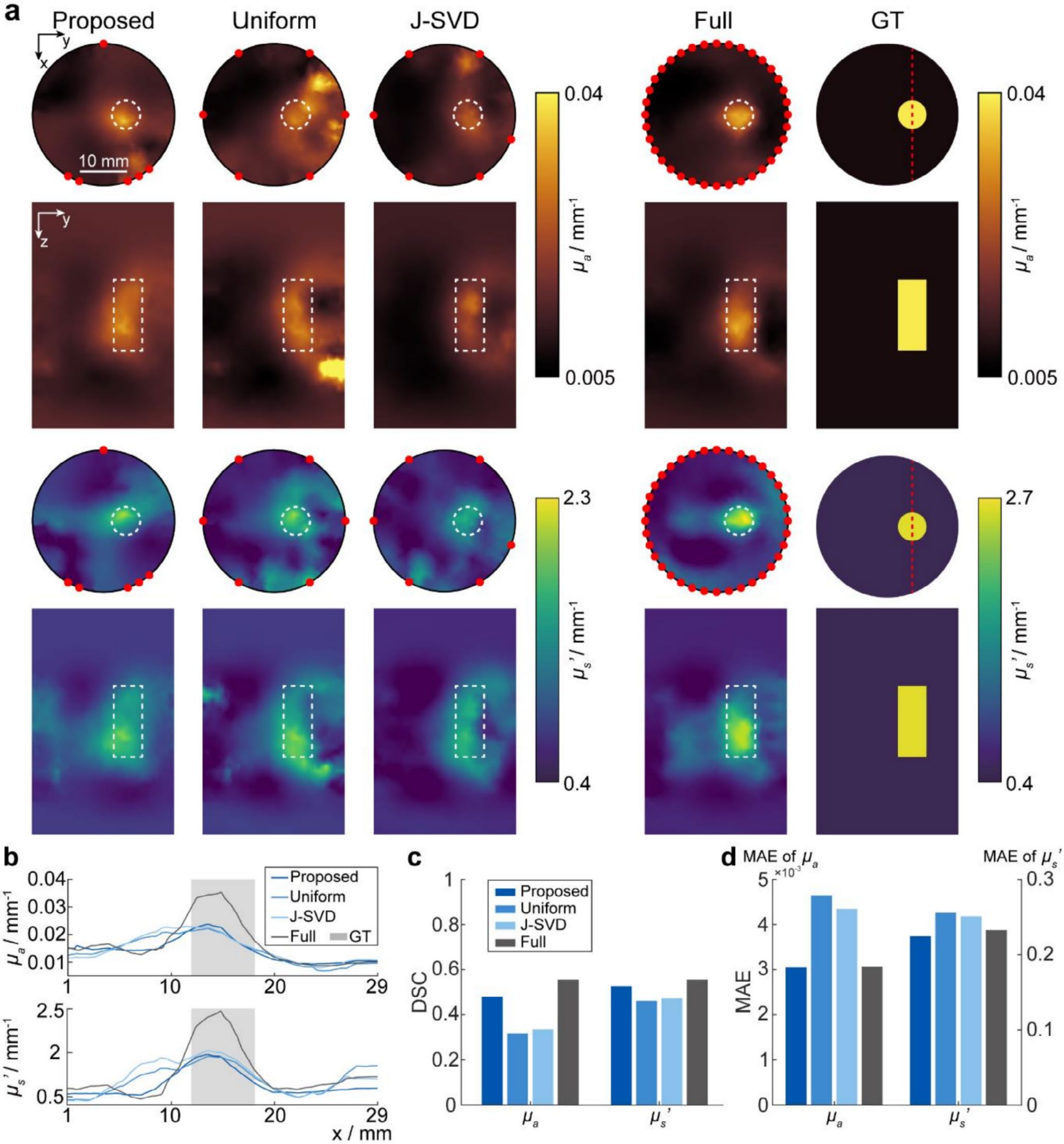

FIGURE 3 | Reconstruction results from the static phantom experiment. (a) Reconstructed images of absorption (μ_a) and scattering (μ_s') coefficients at the *xy*-plane slice at $z = 26$ mm and the *yz*-plane slice at $x = 15$ mm. The real inclusion boundary is indicated with dashed box, and selected projection views are marked with red dots. (b) Cross-sectional profiles extracted along the red dashed lines shown in (a). The gray shaded regions indicate the location of the inclusion. (c) Quantitative evaluation using DSC scores for μ_a and μ_s'. (d) Quantitative evaluation using MAE for μ_a and μ_s'. Scale bar: 10 mm.

(dark blue) closely tracks the trajectory of the full-view result (gray). This structural superiority is quantified by DSC, where our approach significantly outperforms both comparative strategies (Figure 3c). The MAE analysis further shows that our approach achieves the lowest error for both absorption and scattering coefficients, demonstrating its quantitative accuracy in parameter recovery (Figure 3d).

3.3 | Dynamic Imaging Experiment

We further performed dynamic imaging on Phantom B (Figure 4a) to monitor the continuous change of optical properties after contrast agent injection. The corresponding 3D volumetric reconstructions before and after the injection are shown in Figure 4b. The reconstructed slices at $y = -5$ mm and $z = 31$ mm reveal the

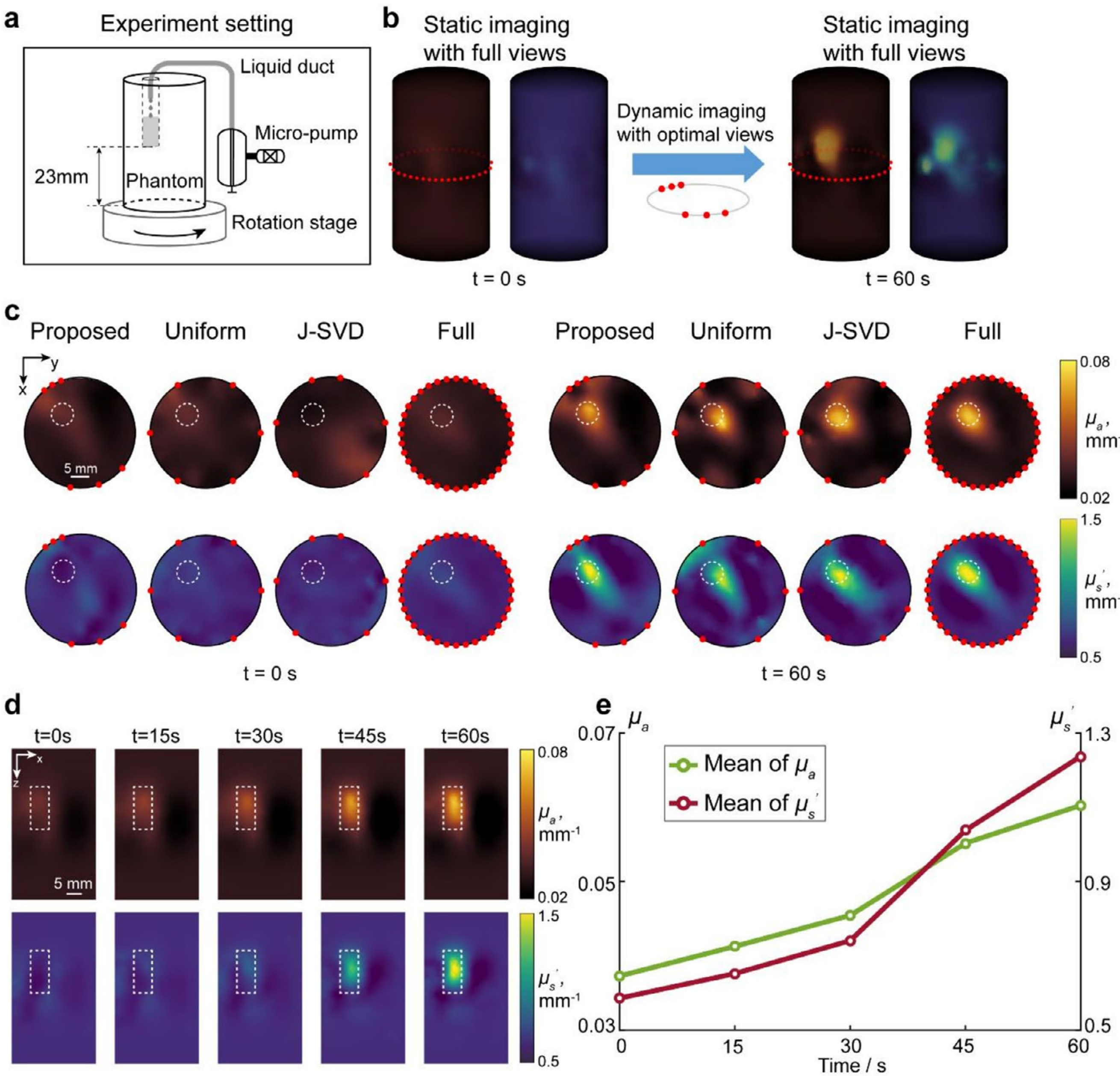

FIGURE 4 | Reconstruction results of dynamic phantom experiment using different view optimization methods. (a) The experimental setup for the dynamic DOT experiment using a micro-pump to continuously inject a high-contrast agent. (b) Volumetric reconstruction of μ_a and μ_s' before and after the injection based on different view optimization methods. (c) Comparison of the reconstructed slice at $z=27$ mm for μ_a and μ_s'. (d) Reconstructed results at different time points in the *xz*-plane slice at $y = 5$ mm. (e) The variation of the mean values of optical properties within the inclusion region. The dashed box indicates the real location of the inclusion. Scale bar: 5 mm.

dynamic changes during the injection process. Figure 4d shows the changes in absorption and scattering coefficients of the dynamically changing inclusion as solution B was constantly added. We tracked the changes in μ_a and μ_s' within a 6 mm-diameter cylindrical region extending from $z = 24$ to 38 mm (Figure 4e). The average value of μ_a increases from 0.037 to 0.096 mm^{-1}, whereas the average value of μ_s' rises from 0.59 to 1.58 mm^{-1}. The linearly increasing trend corresponds well to the injection of solution B at a constant speed in reality. An additional dataset was captured with 36 full views after the injection. Figure 4c presents the comparative results at the $z = 31$ mm cross-section. Consistent with the findings of the previous static experiment, the result obtained by our proposed method aligns well with the full-view result, successfully recovering the shape and location of the high-contrast inclusion. In contrast, uniform sampling introduces boundary artifacts and noticeable spatial displacements in both absorption and scattering maps. Similarly, J-SVD suffers from background fluctuations that obscure the target boundaries. The results show that our view optimization strategy accelerates data acquisition for dynamic monitoring without compromising the reconstruction quality.

4 | Discussion and Conclusion

TD-DOT offers the most comprehensive measurement information, providing the potential of a superior image quality in terms of spatial resolution and depth accuracy. The integration of the emerging SPAD technology into TD-DOT overcomes the limitations of early fiber-based configurations with a limited number of source-detector pairs, offering an increasing number of detection channels. This advancement facilitates the use of free-space TD-DOT for more flexible imaging of small animals and of human being potentially [30]. However, the enriched measurement data also increase computational cost for reconstruction and extended data acquisition time. In this work, we proposed a novel method for optimizing projection views and demonstrated the utility in our homebuilt SPAD-based TD-DOT system. We analyzed the correlation and deviation of the first moment of each pixel sensor derived from the TD measurement, significantly reducing the number of the required views without sacrificing image quality. The results of numerical simulations revealed that our method possesses distinct advantages over uniformly down-sampling and J-SVD strategies. Particularly in complex scenarios involving multiple targets (Case 2) and μ_a/μ_s' interference (Case 3), our approach successfully resolved structural details and decoupled absorption from scattering perturbations. Furthermore, a detailed performance assessment under different contrast levels is provided in Section 2 of Supporting Information. The phantom experiments demonstrate that the proposed methods achieved structural fidelity comparable to the full-view benchmark for stationary targets (static experiment) and successfully captured the changes of optical properties after contrast agent injection (dynamic experiment). Crucially, our method bypasses the expensive methods based on the Jacobian matrix [15–18] in speed, reducing the time from ~300 (J-SVD) to below 1 s. Overall, our data-driven approach, leveraging the statistical features of the time-resolved measurement in the SPAD array readout, is straightforward, rapid and robust for optimizing views and improving the spatiotemporal resolution of TD-DOT.

While our study establishes a robust framework for view optimization in the complicated TD-DOT for the first time, several limitations in the experimental design and algorithmic implementation still exist. Firstly, current studies only involve relatively simple objects, such as cylinder and pillar with homogeneous background. Complex irregular shapes and optical heterogeneity of the imaging objects should be tested for real biological applications in future. Secondly, TD-DOT reconstruction is highly challenging for a large number of time-revolved detection channels, and the reconstructed image quality is also dependent on the inversion algorithm [31]. While transforming measurement data into the FD ensures computational efficiency, exploiting the full temporal profile could further improve reconstruction quality. Strategies such as full time-resolved inversion theoretically maximize information utilization [32]. Alternatively, time-window [33], data domain transforms [34, 35], and usage of temporal moments [36] can also effectively balance reconstruction accuracy and computational cost. Furthermore, full TD features can also be explored for view optimization. While higher-order temporal moments (e.g., variance and skewness) theoretically encode deeper optical variations, our simulations in Section 3 of Supporting Information demonstrate that they are highly sensitive to noise, which severely corrupts the similarity matrix. The dimensional reduction of the forward model [37, 38] and acceleration of inversion procedure [39, 40] will be important for speeding up TD-DOT reconstruction, beyond view optimization. Thirdly, we intend to use a larger SPAD array, which can capture more data, and integrate advanced reconstruction algorithms to achieve super-resolution TD-DOT, as demonstrated in CW-DOT [41] and fluorescence DOT settings [42, 43]. Fourthly, our current dynamic imaging protocol relies on a static subset of views optimized prior to the dynamic event. However, for prolonged continuous monitoring where the target region may experience significant spatial shifts, this initially optimized view configuration might become suboptimal. To address this limitation, future developments could incorporate an adaptive scanning strategy [44, 45]. By interleaving rapid full-view baseline measurements at predefined intervals, the system could dynamically update the statistical matrices and recalibrate the optimal projection views. Last but not least, on the application sides, we plan to translate this data-driven view optimization to functional brain imaging [46, 47] and small animal imaging [48, 49]. For small animal imaging in particular, the proposed TD-DOT framework can be directly combined with continuous-wave fluorescence diffuse optical tomography (CW-FDOT). By providing highly accurate background optical property maps, TD-DOT can significantly improve the fluorescence reconstruction accuracy [50]. Furthermore, we hypothesize that the core concept of our view selection method could be extended to TD-FDOT. Instead of relying on the first temporal moment, the optimization algorithm could evaluate fluorescence-specific temporal features. Metrics such as the peak time, full width at half maximum [51], Fourier-transformed components [52], and Laplace-transformed components [35] have already been proven effective for TD-FDOT image reconstruction. By utilizing these features to compute the similarity and deviation matrices, it is potentially feasible to optimize projection views for TD-FDOT systems, thereby facilitating rapid data acquisition for capturing biological changes.

In conclusion, we developed a fast and effective view optimization method for TD-DOT, without sacrificing the image quality. Both optical absorption and scattering coefficients can be accurately reconstructed with a reduced number of views, and the procedure of data acquisition is significantly shortened. A series of comprehensive numerical and phantom studies validate the reliability and practicality of our method, underscoring the value of our method in developing new-generation TD-DOT systems and many applications that require fast recording of any biological dynamic events, such as brain hemodynamics.

Author Contributions

Wuwei Ren and Edoardo Charbon conceived the study. Linlin Li, Jianru Zhang, and Jiahua Jiang developed the codes and algorithm. Linlin Li, Kaiqi Kuang, and Yang Lin developed the experimental setup. Baile Chen, Jingjing Jiang, and Claudio Bruschini provide guidance for the SPAD-array application. Wuwei Ren supervised the project. All authors participated in writing the manuscript.

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Conflicts of Interest

The authors would like to disclose that Edoardo Charbon is a co-founder of NovoViz. This company has not been involved in the experiments and in the paper drafting, and at the time of the writing has no commercial interests related to this article. The other authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Comparison of reconstruction results using different β across two simulation cases. **Figure S2:** Robustness comparison under varying absorption contrasts. **Figure S3:** Metrics map used to optimize projection views. **Figure S4:** Comparison of reconstruction results using different view optimization metrics across three simulation cases.